

Explainability Requirements for Fitness Routine Recommendations: A Field Case Study in Ecuador

Erick Adalberto Alvia Villegas
Erick Jhair Mera Arias
Alex José Mora Duarte
Mery Helenmey Ponce Rivera
David Octavio Vaca Romero

Facultad de Ciencias de la Computación, Universidad Técnica Estatal de Quevedo
Quevedo, Los Ríos, Ecuador

Abstract

Context: Explainability is increasingly treated as a software-quality concern, yet small organizations often need concrete requirements before an intelligent component exists. **Objective:** We investigate which explainability requirements emerge for a proposed fitness-routine recommender in FabroGym, a local-gym management system, and how those requirements change after member checking. **Method:** We conducted a bounded requirements-engineering case study using ten initial interviews and six prototype walkthroughs (three technical and three non-technical), complemented by a 70-response client questionnaire. The walkthrough corpus contains 76 coded fragments and 37 normalized codes. Explainability-relevant fragments were transformed into candidate non-functional requirements and reviewed by three prior participants (MC-P01–MC-P03). All numerical outputs are reproduced by versioned Python scripts. **Results:** Nine fragments were pertinent to explainability, seven from technical and two from non-technical walkthroughs. They produced four candidate requirements concerning recommendation rationale and constraints, human review/provenance, routine history, and the principal criterion behind a suggestion or automatic assignment. Member checking yielded 12 decisions: four confirmations and eight adjustments; all four candidates were retained with changes. At the independent-session level (3 technical vs. 3 non-technical), Cliff’s delta for the proportion of explainability-relevant fragments was 0.556 with an exact-bootstrap 95% CI of [-0.333, 1.000]; the estimate is reported as descriptive and not population-interpretable. The strict code-saturation indicator for the final three walkthroughs was 6.306%, above the predefined 5% threshold, while axial-category stabilization was 1.852%. **Conclusions:** Explainability requirements can be elicited and operationalized before implementing a recommender, but the evidence supports specification and validation rather than claims about an implemented AI component or generalized user effects.

Keywords: Explainability; Non-functional requirements; Fitness recommendation; Requirements engineering; Field study

1 Introduction

Small fitness centers coordinate memberships, payments, attendance, product sales, instructor availability, training routines, and day-to-day handovers under strong time and staffing constraints. In such settings, requirements engineering (RE) is not merely a documentation exercise: it is the mechanism used to separate observed operational needs from attractive but unsupported functionality. The FabroGym project was developed around that distinction. Its terminal requirements corpus follows ISO/IEC/IEEE 29148:2018 and the quality model of ISO/IEC 25010:2023, while the empirical component

is treated as a bounded software-engineering case study ISO/IEC/IEEE [2018], ISO/IEC [2023], Pohl [2010], Washizaki et al. [2024], Runeson and Höst [2009], Wohlin et al. [2012].

The research problem becomes sharper when a future intelligent component is considered. Recommendation systems can provide personalized alternatives, but a useful recommendation is not automatically an understandable one. Decades of recommender-system research show that explanations may affect transparency, acceptance, trust, and decision quality Herlocker et al. [2000], Gedikli et al. [2014], Zhang and Chen [2020]. Explainable-AI research likewise emphasizes that explanation is contextual, audience-dependent, and shaped by human expectations Miller [2019], Ribeiro et al. [2016], Sokol and Flach [2020]. From an RE perspective, this means that explainability should be specified as a quality requirement before selecting or deploying an explanation technique. Chazette and colleagues explicitly frame explainability as a non-functional requirement (NFR) and provide models for connecting explanation needs with software-quality concerns Chazette and Schneider [2020], Chazette et al. [2021, 2022].

This study addresses a practical gap between those conceptual frameworks and an early-stage, real-world academic system in which the recommender itself is only proposed. FabroGym’s operational MVP manages gym workflows with synthetic demonstration data; it does not contain a machine-learning recommender. Consequently, this paper does not evaluate prediction accuracy, recommendation quality, or an implemented explanation interface. Instead, it asks what should be explainable if such a component is later built, and how stakeholder evidence can be transformed into measurable requirements without inventing data or prematurely fixing an algorithm.

The contribution is fourfold. First, we provide a traceable evidence-to-requirement pipeline that preserves the distinction between ten initial interviews and six later walk-through sessions. Second, we report a technical/non-technical comparison at the level supported by the actual data, without creating inferential outcomes that were not collected. Third, we operationalize four explainability NFRs with a metric, threshold, and verification method and report how three prior participants changed their wording through member checking. Fourth, we publish the reproducible descriptive analysis and explicitly report deviations from the OSF protocol rather than retrofitting the study after observing results Nosek et al. [2018], Simmons et al. [2011], Kerr [1998].

The study answers two research questions. **RQ1: What explainability information needs emerge from technical and non-technical walkthrough evidence for a proposed fitness-routine recommender?** **RQ2: How are those needs transformed into verifiable explainability NFRs, and how are the candidate requirements modified through member checking?** These RQs deliberately focus on requirement discovery and validation. They do not assume that an intelligent component exists, and they do not claim statistical generalization beyond the case.

2 Related work

2.1 Requirements engineering and evidence traceability

RE has long been characterized as the process of discovering, analyzing, documenting, validating, and evolving stakeholder needs Nuseibeh and Easterbrook [2000], van Lamsweerde [2001], Pohl [2010]. ISO/IEC/IEEE 29148 emphasizes properties such as necessity, unambiguity, completeness, feasibility, verifiability, and traceability, while Glinz highlights the continuing difficulty of precisely defining and handling NFRs ISO/IEC/IEEE [2018], Glinz [2007]. Empirical studies also show that industrial RE problems remain entangled with communication, incomplete knowledge, and context Mendez Fernandez et al. [2017], Ferrari et al. [2016]. FabroGym therefore treats each requirement as an evidence-linked statement rather than a free-standing design idea.

Traceability also supports acceptance and reproducibility. Automatically deriving acceptance tests from natural-language conditions illustrates how explicit requirement structure can support verification Fischbach et al. [2023]. Reproducible computational research, in turn, requires analysis steps that can be regenerated from versioned inputs rather than manually edited tables Sandve et al. [2013], Peng [2011]. Software and source-code citation principles extend that reproducibility chain beyond prose documents

Smith et al. [2016], Di Cosmo et al. [2020].

2.2 Explainability as a requirement

Chazette and Schneider identify explainability as an emerging NFR whose meaning depends on stakeholders, the object being explained, the explanation purpose, and interactions with other quality properties Chazette and Schneider [2020]. The subsequent explainability model and knowledge catalogue systematize those concepts Chazette et al. [2021], and later work connects requirements analysis to evaluation of explainable software systems Chazette et al. [2022]. These works motivate our decision to formulate explanation obligations before implementation and to express them with observable criteria rather than generic claims such as “the recommender shall be transparent.”

The broader XAI literature is complementary but not interchangeable with RE. Miller argues that good explanations are selective and social rather than exhaustive technical dumps Miller [2019]. LIME demonstrates post-hoc local explanations for classifiers Ribeiro et al. [2016], whereas explainability fact sheets provide a structured way to describe explanation approaches Sokol and Flach [2020]. FabroGym does not select among these techniques: the artifact under study is the requirement, not the explanation algorithm.

2.3 Explanations in recommender systems

Recommender-system studies provide domain-relevant evidence about why explanations matter. Herlocker et al. showed that explanation interfaces can increase acceptance of collaborative-filtering recommendations Herlocker et al. [2000]. Gedikli et al. compared multiple explanation types and found that explanation design changes perceived transparency and satisfaction Gedikli et al. [2014]. Zhang and Chen synthesize explainable recommendation into a broader taxonomy covering explanation generation, users, and evaluation goals Zhang and Chen [2020]. These studies evaluate built recommendation mechanisms; our case instead uses stakeholder interactions to define requirements for a component that is explicitly not implemented.

2.4 Case-study validity, qualitative saturation, and member checking

Software-engineering case studies require clear case boundaries, data sources, chains of evidence, and explicit threats to validity Runeson and Höst [2009], Siegmund et al. [2015]. Interview-based work additionally requires disciplined handling of saturation. Guest et al. and Hagaman and Wutich illustrate how the number of interviews needed depends on the type of theme sought, while Hennink et al. distinguish code saturation from meaning saturation Guest et al. [2006], Hagaman and Wutich [2017], Hennink et al. [2017]. We therefore report the project’s prespecified code-level indicator numerically and do not rename axial stabilization as code saturation.

Member checking can strengthen interpretive credibility, but it is not a mechanical guarantee of truth Birt et al. [2016]. In this study it is used narrowly: prior participants review four candidate NFRs and may confirm, adjust, or reject each interpretation. This creates an auditable chain from evidence fragment to candidate requirement to participant decision to final requirement.

2.5 Closed related-work screening

The closed search for this manuscript used the 29-entry bibliography already versioned with the 2A manuscript as the initial candidate pool. All 29 entries were screened for direct relevance to (i) explainability requirements, (ii) recommender explanations, (iii) requirements elicitation/traceability, and (iv) qualitative validation. Twelve closest studies were retained for the comparative synthesis in Table 1; methodological and normative sources were retained elsewhere in the bibliography. This procedure follows the principle of documenting search terms, inclusion/exclusion logic, and the screened

set rather than inventing database hit counts that were not recorded Kitchenham and Charters [2007].

Table 1: Closest related studies and the specific distinction of the FabroGym case.

Reference	Year	Focus	Main result / role	Difference in this study
Chazette & Schneider Chazette and Schneider [2020]	2020	Explainability NFR	Challenges and recommendations for specifying explainability	Operationalizes four field-derived RNF for a proposed local-gym recommender.
Chazette et al. Chazette et al. [2021]	2021	Explainability model	Definition, model and knowledge catalogue	Uses observed walkthrough evidence and participant review to instantiate requirements.
Chazette et al. Chazette et al. [2022]	2022	RE-to-evaluation	Connects requirements analysis to system evaluation	Works before implementation and limits claims to specification/validation.
Miller Miller [2019]	2019	Human explanation	Social-science properties of useful explanations	Converts explanation needs into verifiable software requirements.
Zhang & Chen Zhang and Chen [2020]	2020	Explainable recommendation	Taxonomy of explainable recommender research	Applies requirements elicitation to fitness-routine recommendations.
Gedikli et al. Gedikli et al. [2014]	2014	Explanation interfaces	Explanation types affect perceived quality	Elicits required information before building explanation interfaces.
Herlocker et al. Herlocker et al. [2000]	2000	Collaborative filtering	Explanations can support acceptance/transparency	Does not evaluate an implemented recommender.
Sokol & Flach Sokol and Flach [2020]	2020	XAI assessment	Structured explainability assessment	Defines verification targets without choosing an explainer.
Ferrari et al. Ferrari et al. [2016]	2016	Elicitation interviews	Tacit knowledge and ambiguity shape elicitation	Combines interview heritage with prototype walkthrough evidence.
Runeson & Höst Runeson and Höst [2009]	2009	Case-study method	Reporting guidance for SE case studies	Applies a bounded Ecuadorian academic case with explicit deviations.
Birt et al. Birt et al. [2016]	2016	Member checking	Benefits and caveats of member validation	Uses decision-level review of candidate RNF.
Hennink et al. Hennink et al. [2017]	2017	Saturation	Distinguishes code and meaning saturation	Reports the 5% rule outcome without overstating saturation.

3 Methodology

3.1 Study design and case boundary

We used a single-case, mixed-evidence RE design. The case is FabroGym, an academic system grounded in the operations of a local gym in Ecuador. The empirical focus of this paper is explainability as an NFR for a *proposed* routine-recommendation capability. The operational MVP and the proposed recommender are intentionally separated: the first supports routine management and other gym workflows, whereas the latter is a future component whose explanation obligations can be specified before implementation.

The study follows software-engineering case-study guidance Runeson and Höst [2009], Wohlin et al. [2012], Jedlitschka et al. [2008]. Survey handling is informed by Moller et al. [2020]; thematic saturation is interpreted using Guest et al., Hagaman and Wutich, and Hennink et al. [2006], Hagaman and Wutich [2017], Hennink et al. [2017]; and member checking follows the role-aware caution described by Birt et al. [2016]. Reproducible analyses are implemented in Python, with generated tables and figures versioned together with their data inputs Sandve et al. [2013], Peng

[2011].

3.2 Research questions in PICOC form

For **RQ1**, the *Population* comprises technical and non-technical participants interacting with the FabroGym prototype; the *Intervention* is the structured walkthrough and thematic coding; the *Comparison* is technical versus non-technical profile evidence at a descriptive level; the *Outcome* is the set of explainability-relevant fragments, labels, and information needs; and the *Context* is a local-gym management system with a proposed routine recommender.

For **RQ2**, the *Population* comprises three previous participants invited to member checking; the *Intervention* is review of four candidate explainability NFRs; the *Comparison* is candidate wording versus participant-confirmed or adjusted wording; the *Outcome* is the final set of measurable RNF and the distribution of confirm/adjust/reject decisions; and the *Context* is the same proposed recommendation capability.

No inferential hypothesis was preregistered for the three-versus-three walkthrough profile comparison. The operational propositions were instead: **P1**, technical and non-technical walkthroughs may surface partially shared but differently emphasized quality concerns; and **P2**, participant review will either confirm, adjust, or reject each candidate explainability requirement, producing a traceable final decision.

3.3 Participants and evidence sources

The accumulated audiovisual package contains 16 sessions: ten initial interviews (ENTR-01 to ENTR-10), three technical walkthroughs (WALK-TEC-01 to WALK-TEC-03), and three non-technical walkthroughs (WALK-NTEC-01 to WALK-NTEC-03). The identifiers are preserved across fichas, transcripts, analysis, and traceability; WALK sessions are not renamed as interviews. The public technical sheet records 16 videos totaling 06:18:08 and 16 audios totaling 06:18:14. To avoid double counting, the video duration is used as the one-session duration for the terminal 240-minute criterion.

A client questionnaire contains 70 accepted responses. Its two direct-identification columns are empty in all 70 rows. Importantly, the instrument has neither a technical/non-technical profile field nor an explainability Likert scale. It is therefore used only to describe general domain perceptions. The member-checking record contains three pseudonymized prior participants, MC-P01, MC-P02, and MC-P03, reviewing four candidate requirements on 29 August 2026.

Table 2: Empirical sources used in the terminal analysis.

Source	N	Analytical role
Initial interviews (ENTR)	10	Base operational requirements, constraints, domain context.
Technical walkthroughs	3	Prototype validation, coding, explainability evidence.
Non-technical walkthroughs	3	Prototype validation, coding, explainability evidence.
Client questionnaire	70	Descriptive domain evidence only.
Member checking	3	Review of four RNF candidates, 12 nominal decisions.

3.4 Thematic coding and explainability extraction

The six walkthroughs produced 76 coded fragments, of which 49 came from technical sessions and 27 from non-technical sessions. Open coding was normalized into 37 distinct codes and 18 thematic categories. Each fragment retains the session identifier, profile, source text, category, normalized code, and an explainability-applicability flag. The explainability subset contains nine fragments: seven technical and two non-technical.

Explainability labels were multi-valued because one fragment could express, for example, both justification and context. Consequently, label frequencies are descriptive and do not define a closed partition. We do not report “framework coverage” as a percentage because no preregistered denominator maps every possible framework dimension to a mutually

exclusive coding scheme. Doing so would create a numerical result not supported by the instrument.

3.5 From evidence to candidate and final NFRs

For each explainability-relevant cluster, we constructed a candidate NFR using the structure: object to explain, audience, presentation format, moment of presentation, metric, threshold, verification method, and source fragments. This operationalization is consistent with treating NFRs as testable quality obligations rather than vague aspirations Glinz [2007], Chazette and Schneider [2020], Chazette et al. [2021].

The four candidates were then presented to MC-P01–MC-P03. Each participant recorded one of three decisions: *Confirmed*, *Adjusted*, or *Not confirmed*, with a rationale and replacement wording when adjustment was requested. The final requirement was produced only after considering all three decisions; no candidate was silently promoted to the final specification.

3.6 Questionnaire analysis and statistical scope

The questionnaire is analyzed through category frequencies and six ordinal 1–5 indices (expiry difficulty, payment wait, registration ease, plan clarity, staff satisfaction, and privacy importance). For the ordinal means, 95% bootstrap intervals use 10,000 resamples with a fixed random seed in the versioned script. These indices summarize the questionnaire only; they are not reinterpreted as explainability satisfaction scores.

The analysis explicitly does *not* compute Fleiss’ κ , Mann–Whitney U , Shapiro–Wilk, or Levene tests. Fleiss’ κ would require comparable ratings of the same items by multiple raters rather than the available member-checking decision process Cohen [1960], Fleiss [1971]. Mann–Whitney would require a preregistered quantitative outcome at an appropriate independent unit. The terminal correction derives a session-level descriptive outcome (the proportion of explainability-relevant fragments) only for effect-size reporting; it is not retroactively treated as a preregistered participant-level inferential endpoint. Normality and variance tests would therefore not answer a meaningful preregistered hypothesis. This non-applicability is recorded rather than filling the manuscript with unsupported p-values.

For the technical/non-technical contrast, the independent unit is the WALK session rather than a thematic category. There are six independent units in total (three per profile). We report Cliff’s delta on the per-session proportion of explainability-relevant fragments and an exact bootstrap percentile interval obtained by enumerating all ordered with-replacement resamples of size three within each profile ($27 \times 27 = 729$ paired bootstrap combinations) Cliff [1993]. The rubric-facing field **interpretable** is defined narrowly as population/inferential interpretability. It is set to NO because only three independent sessions exist per profile; the estimate remains valid as a descriptive-exploratory summary of this case. A secondary delta on raw counts is retained only as a sensitivity analysis because the total number of coded fragments differs across sessions.

For sample-size context, a reproducible Cohen $d = 0.5$, $\alpha = .05$, power = .80 calculation yields approximately 34 observations for a one-sample/paired standardized effect and approximately 64 per group for two equal independent groups Cohen [1988]. The questionnaire $n = 70$ exceeds the first number but cannot support a technical-versus-non-technical contrast because it contains no profile variable.

3.7 Saturation rule

The terminal rubric operationalizes code saturation as the average number of new codes in the last three sessions being at most 5% of the final accumulated code set. With 37 normalized codes and 2.333 new codes per session on average across WALK-NTEC-01, WALK-NTEC-02, and WALK-NTEC-03, the observed value is 6.306%. It therefore does not satisfy the strict code-level threshold. The axial-category analysis accumulates 18 categories and yields 1.852% for the same three-session window; this is reported as complementary conceptual stabilization, not as a substitute for the code rule Hennink

et al. [2017].

3.8 Ethics, privacy, and open-science controls

Public artifacts are anonymized or pseudonymized; identifiable voices, faces, signatures, and original consent material belong to a restricted encrypted zone. The public analysis uses session codes rather than names. These controls align the project with Ecuador’s Organic Law on Personal Data Protection and privacy-by-design guidance Asamblea Nacional del Ecuador [2021], Superintendencia de Protección de Datos Personales [2025]. The software requirements additionally reference access-control, session, and accessibility baselines from OWASP ASVS, NIST digital-identity guidance, and WCAG 2.2 OWASP Foundation [2025], Temoshok et al. [2025], W3C [2024].

The OSF protocol (v1.4; DOI 10.17605/OSF.IO/62YSC) was registered on 29 August 2026. The six WALK sessions occurred before that timestamp. We therefore classify them transparently as pre-registration/formative evidence rather than pretending they were collected after preregistration. The deviation is documented in the OSF deviations artifact, consistent with the principle that preregistration should expose rather than conceal deviations Nosek et al. [2018], Simmons et al. [2011], Kerr [1998].

Closure correction to consent evidence. The closure review identified four public consent copies (ENTR-02, ENTR-03, ENTR-04, and ENTR-06) whose prior PDF representation originated from editable Word documents and therefore did not itself demonstrate a scanned signature. Those four public copies were replaced by rasterized, redacted scans of the signed forms while preserving the session code, date, authorization structure, and public/restricted separation. In the current public layer, `pdftotext` returns zero words for each of those four files, which is consistent with their rasterized scan representation rather than editable Word-origin text. The control record is maintained in `08_Etica/CONTROL_CONSENTIMIENTOS_FINAL.md` and the metadata audit in `07_Datos/resultados/tablas/B6_control_metadatos_consentimientos.csv`. This correction changes the evidentiary representation of the consent proof; it does not alter participant codes, session chronology, or the empirical dataset.

The reproducible session inventory contains 16 individual consent forms: 10 interview consents, 3 technical-walkthrough consents, and 3 non-technical-walkthrough consents, matching the 16 documented field sessions. The closure guide mentions 17 consents, but no seventeenth session-specific form is identifiable in the reproducible evidence without duplicating or reclassifying another document. The discrepancy is therefore documented rather than artificially filled. The canonical public copies remain in `02_Evidencias/Consentimientos/`; a byte-identical mirror in `08_Etica/consentimientos/` exists only so that the guide’s literal verification path can be executed.

4 Results

4.1 RQ1: Explainability information needs

The six walkthroughs yielded 76 coded fragments: 49 technical and 27 non-technical. The strongest shared categories were usability/navigation, routines/personalization, clients/memberships/payments, clarity/information, and assistance/automation. Technical sessions additionally contained exclusive or predominant references to communication, maintainability/scalability, legal/privacy, security/continuity, security/operational traceability, data validation, and validation engineering. Non-technical sessions uniquely emphasized prevention and recovery from errors. Figure 1 summarizes the ten highest-frequency categories. These counts describe the coded corpus; they are not treated as independent participant measurements.

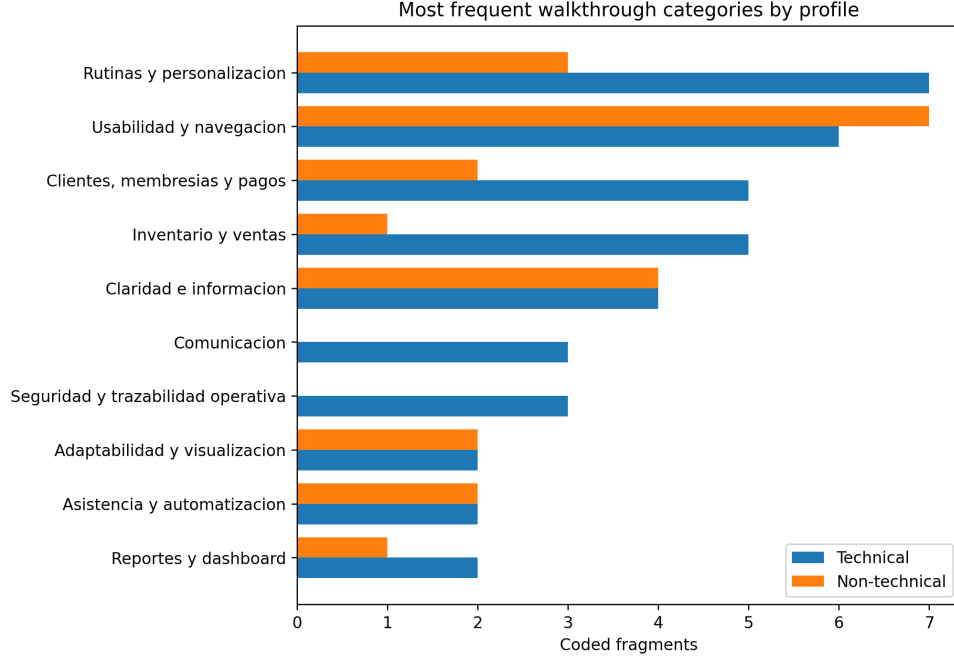


Figure 1: Most frequent walkthrough categories by technical and non-technical profile. Counts are coded fragments, not inferential observations.

Nine fragments were explicitly pertinent to explainability, seven from technical walkthroughs and two from non-technical walkthroughs. Multi-label coding most often identified *justification* (4 fragments) and *transparency* (3), followed by provenance, personalization, and context (2 each). Comprehensibility, accessibility, detail, validation, and logic appeared once each. Since one fragment may receive multiple labels, these frequencies do not sum to nine and are not percentages of a closed framework.

At the session level, the technical proportions of explainability-relevant fragments were 2/15 (0.133), 4/16 (0.250), and 1/18 (0.056); the non-technical proportions were 1/8 (0.125), 0/8 (0.000), and 1/11 (0.091). The primary Cliff's delta was 0.555556 with an exact-bootstrap 95% CI of [-0.333333, 1.000000] across $n_{unidades}=6$ independent sessions. The terminal table marks `interpretable=NO` for inferential/population interpretation because there are only three sessions per profile and the interval is wide and includes zero. A raw-count sensitivity analysis produced $\delta=0.777778$ and 95% CI [0.333333, 1.000000], but it is likewise not interpreted inferentially because session fragment totals differ. These effect-size outputs describe the observed case; they do not convert the 18 thematic categories into independent observations.

Table 3: Session-level technical vs. non-technical effect-size audit.

Metric	n_{units}	Cliff's δ	Exact-bootstrap 95% CI	Interpretable ^a
Explainability-relevant proportion per WALK session	6 (3+3)	0.555556	[-0.333333, 1.000000]	NO
Raw explainability-relevant count per WALK session (sensitivity)	6 (3+3)	0.777778	[0.333333, 1.000000]	NO

^aNO means not interpretable as a population/inferential effect. Both estimates remain descriptive summaries of the six observed sessions; the raw-count row is secondary because session fragment totals differ.

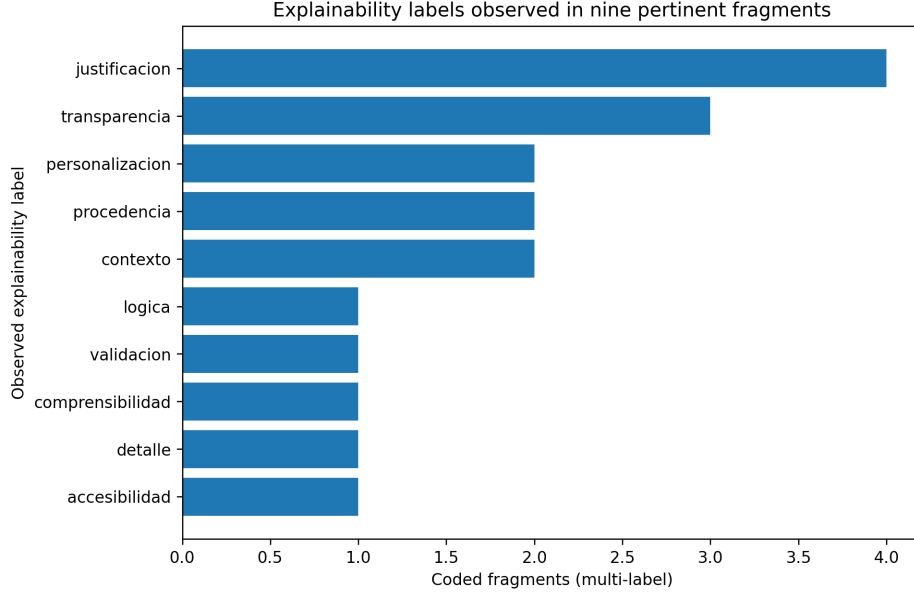


Figure 2: Observed explainability labels in the nine pertinent walkthrough fragments. Multi-label coding is allowed.

The evidence clusters converged on four information needs. First, a recommendation should expose its primary objective or need and relevant recorded constraints. Second, users should be able to see whether a routine has been reviewed by an authorized trainer and, when applicable, the provenance of an external professional indication. Third, the system should preserve an inspectable routine history with dates and available observations. Fourth, any future suggestion or automatic assignment should state the principal criterion used. These needs are intentionally system- and process-facing; they do not prescribe LIME, SHAP, feature importance, counterfactuals, or another explanation algorithm Miller [2019], Ribeiro et al. [2016], Sokol and Flach [2020].

The client questionnaire provides complementary domain context but not explainability measurement. Privacy importance had an ordinal mean of 3.971 on the 1–5 coding (95% bootstrap CI 3.700–4.214); 54 of 70 respondents (77.143%) selected “Important” or “Very important.” The most frequently selected first improvement was memberships (15/70, 21.429%), followed by communication/notifications and customer attention (14/70 each, 20.000%). These results support the relevance of transparent data handling and clear operational information, but they are not transformed into an explainability satisfaction score.

4.2 RQ2: Member checking and terminal NFRs

The four candidate NFRs generated 12 member-checking decisions. Four decisions (33.333%) were confirmations, eight (66.667%) requested adjustments, and none rejected a candidate. RNF-EXP-C01 and RNF-EXP-C02 each received two confirmations and one adjustment. RNF-EXP-C03 and RNF-EXP-C04 each received three adjustments. All four were therefore incorporated with revised wording rather than treated as unanimously accepted originals.

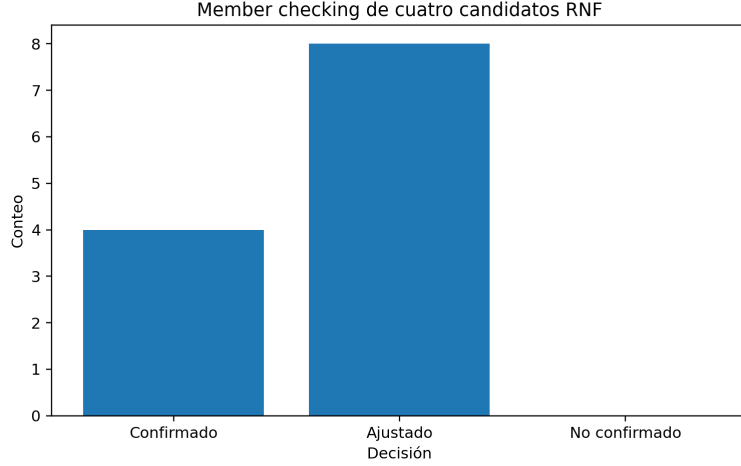


Figure 3: Member-checking decisions across four candidate explainability NFRs (12 decisions).

The most consequential adjustments narrowed overclaims. For RNF-16, the final wording removed a general reference to observations and retained only the objective/need plus relevant recorded constraints. For RNF-17, participants retained the need to show trainer review and professional provenance but avoided inventing states not observed in the evidence. For RNF-18, all three participants removed an automatic comparison between routine versions and kept an inspectable history with dates and available observations. For RNF-19, all three removed the claim that the system must compare alternatives and retained only the principal criterion behind a future suggestion or automatic assignment.

Table 4: Terminal explainability NFRs after member checking.

ID	Terminal requirement	Metric/threshold	Verification
RNF-16	A proposed routine recommendation shall show its primary objective or need and, when present, relevant recorded client constraints.	100% of test recommendations show the objective/need and applicable constraints.	Functional cases with different objectives and constraints; inspect visible routine information.
RNF-17	Every assigned or proposed routine shall indicate whether it was reviewed/validated by an authorized trainer and, when recorded, the provenance of an external professional indication.	100% show review state; cases with external indications show provenance.	Functional cases with reviewed/unreviewed routines and with/without external indications.
RNF-18	The system shall allow consultation of the current routine and the history of registered routines, showing date and associated observations when available.	100% of test routines remain consultable with date; observations appear when present.	Register multiple routines and verify later retrieval, dates, and available observations.
RNF-19	When the system suggests an initial routine or automatically assigns a routine/class, it shall briefly show the principal criterion used for the suggestion or assignment.	100% of evaluated future suggestions/automatic assignments show a principal criterion.	Cases with different profiles/objectives; inspect the criterion shown before confirmation.

All four final RNF include a metric, threshold, and verification method; thus operationalization completeness for this four-requirement set is 100%. This is not equivalent to implementation coverage: the recommendation component remains **proposed**, and

each RNF is classified as **specified**, not implemented in the MVP.

4.3 Saturation and reproducibility checks

The code curve shows 10, 12, 8, 2, 2, and 3 new normalized codes across the six walkthroughs, accumulating to 37. The final three sessions average 2.333 new codes, or 6.306% of the final code set. Therefore, the study does not satisfy its strict $\leq 5\%$ code-level rule. The inflection is nevertheless visible after the third session, and the axial-category curve falls to 1.852% for the final three sessions. Figure 4 presents both new and accumulated codes; Figure 5 presents axial stabilization.

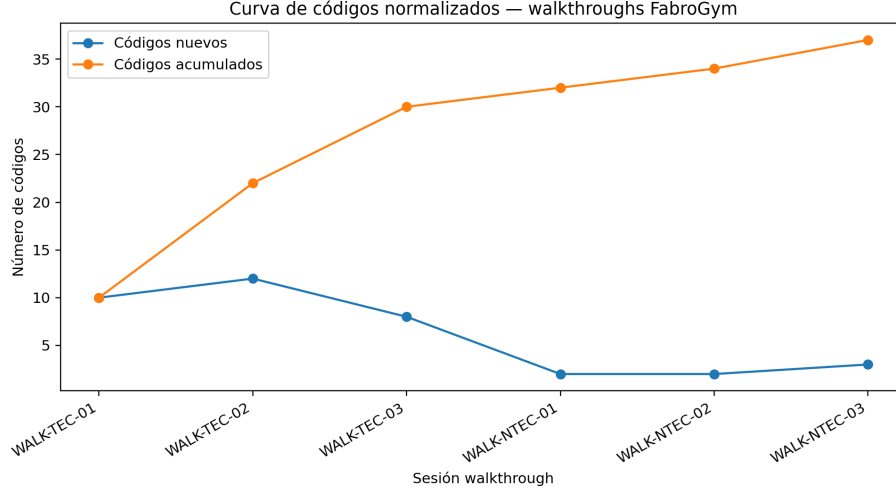


Figure 4: New and accumulated normalized codes across the six walkthrough sessions.

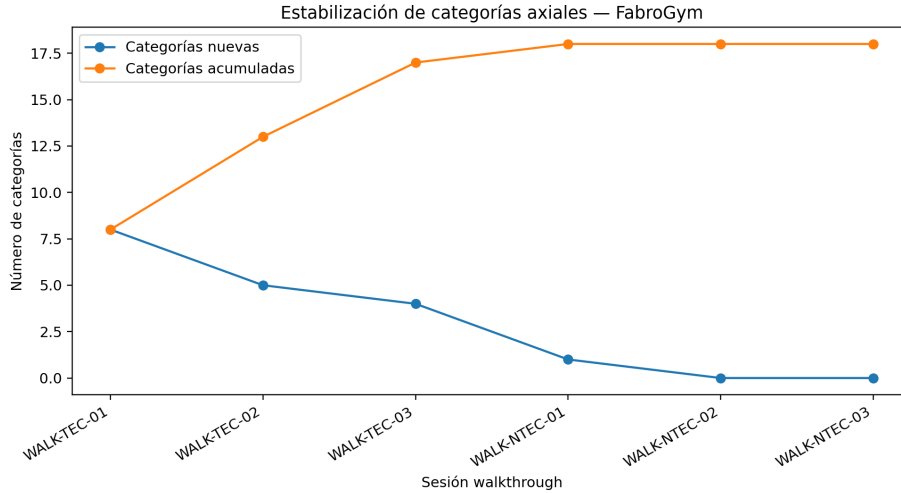


Figure 5: New and accumulated axial categories. Axial stabilization is complementary and does not replace the code-level threshold.

Every numerical statement in this Results section corresponds to a CSV output generated from the versioned analysis package. The single entry point regenerates multimedia summaries, questionnaire frequencies and bootstrap intervals, saturation tables, and member-checking summaries. This design follows reproducible-research principles and reduces the risk of manuscript-only numbers Sandve et al. [2013], Peng [2011], Wilkinson et al. [2016], Smith et al. [2016].

5 Discussion

5.1 Implications for requirements-engineering research

The main research implication is that explainability can be engineered before an intelligent component is implemented. In FabroGym, no participant needed to inspect a machine-learning model to express requirements about rationale, constraints, provenance, history, or assignment criteria. This supports the RE view that explainability is a stakeholder-dependent quality property rather than a synonym for a particular post-hoc algorithm Chazette and Schneider [2020], Chazette et al. [2021, 2022]. It also suggests a sequencing benefit: specifying explanation obligations early can constrain later architecture and evaluation instead of forcing an explanation layer onto a completed black box.

The second implication concerns the granularity of evidence. Technical walkthroughs contributed more explainability-relevant fragments (7 versus 2), but that asymmetry cannot be interpreted as a population effect. Technical sessions also produced more coded fragments overall (49 versus 27). The corrected effect-size analysis uses the six sessions, not the 18 thematic categories, as independent units. Its primary delta is positive (0.556), but the exact-bootstrap 95% interval spans -0.333 to 1.000 and only three sessions exist per profile. Accordingly, the `interpretable` field is `NO` for population inference. The defensible finding is case-descriptive: technical sessions more often articulated provenance, validation, security, and system-oriented rationale, while non-technical evidence helped constrain clarity and avoid overcomplicated comparisons.

The third implication is methodological. Member checking did not merely rubber-stamp the four candidates. Two-thirds of the 12 decisions requested adjustments, and the strongest adjustments removed functionality that the analysts had added beyond the evidence: automatic version comparisons, comparisons against alternatives, and overly broad observation fields. That pattern shows why member checking can be valuable in RE: it can act as a brake on analyst elaboration. At the same time, Birt et al.’s caution remains relevant; participant confirmation is evidence about interpretive fit, not proof that the requirement is universally correct Birt et al. [2016].

5.2 Implications for practice

For a small gym, explanation requirements should remain concise and actionable. A “reason for recommendation” that exposes the primary goal and relevant constraints is likely to be more usable than a technical explanation of model weights. Trainer-review status and professional provenance address accountability: users can distinguish system-originated guidance from human or external professional input. A routine history with dates supports traceability and continuity. Finally, showing the principal criterion behind any future automatic assignment gives both instructors and clients a basis for questioning or overriding the suggestion.

These obligations are compatible with several recommender architectures. A rules-based recommender could satisfy them by exposing matched rules; a content-based or collaborative system could provide a human-readable rationale derived from its inputs; a more complex model might require a separate explanation mechanism. The requirements therefore preserve architectural freedom while establishing acceptance conditions. This is consistent with the goal of NFR specification: define externally verifiable quality outcomes before binding the solution to one implementation Glinz [2007], ISO/IEC [2023].

Privacy is also inseparable from explainability in this domain. An explanation should not become a new channel for exposing sensitive personal attributes. The project therefore minimizes real personal data and excludes health, biometric, weight, body-measure, address, and body-image data from development/testing by default. This aligns with privacy-by-design principles under Ecuadorian law and with general secure-design guidance Asamblea Nacional del Ecuador [2021], Superintendencia de Protección de Datos Personales [2025], OWASP Foundation [2025], Temoshok et al. [2025].

5.3 Counterintuitive and negative findings

Two negative findings are important. First, the 70-response questionnaire cannot answer the paper’s technical-versus-non-technical explainability question. Its size is larger than the qualitative walkthrough set, but sample size cannot compensate for a missing grouping variable or missing explainability items. Recasting general satisfaction questions as explainability Likert items would produce a more impressive statistical section but would violate construct validity.

Second, the code-saturation threshold is not met. The observed 6.306% is close to but above 5%, and the final non-technical session adds three new codes. Reporting axial stabilization at 1.852% does not authorize calling the code-level criterion satisfied. This negative result is informative: the themes appear to be organizing into stable higher-level categories while fine-grained codes continue to emerge. Future data collection could test whether the curve converges with additional walkthroughs.

5.4 Lessons learned

The most transferable lesson is procedural: keep identifiers stable, encode limitations in the analysis, and allow participants to delete analyst-added meaning. The distinction between ENTR and WALK identifiers prevented accidental reclassification of validation sessions as interviews. The separation between a proposed recommender and an implemented MVP prevented a specification study from becoming a fictitious system evaluation. Finally, the decision to produce no Fleiss κ or Mann–Whitney test when the instrument did not support them preserved the chain of evidence even though it reduced the number of statistical outputs.

6 Threats to validity

We organize threats using the standard internal, external, construct, and conclusion categories commonly applied in empirical software engineering Wohlin et al. [2012], Siegmund et al. [2015].

6.1 Internal validity

The first internal threat is temporal: all six walkthroughs occurred before the OSF registration timestamp. This means the sessions cannot be portrayed as confirmatory post-registration data collection. We mitigate the issue through explicit deviation reporting and by keeping the analysis descriptive; the temporal ordering itself cannot be repaired retroactively. The second threat concerns member checking. The activity is documented with pseudonymized decisions and a date, but no audiovisual recording exists. We retain the documentary evidence and explicitly report the missing recording rather than creating a substitute.

6.2 External validity

FabroGym is a single local-gym case and uses non-probability recruitment. Operational practices, technology familiarity, and expectations may differ in larger chains, other countries, or other recommendation domains. We therefore restrict conclusions to the case and use the study to generate verifiable requirements rather than universal prevalence estimates. The six walkthroughs include only three technical and three non-technical sessions. Replication with larger and more diverse samples is needed before treating observed profile patterns as stable population differences.

6.3 Construct validity

The questionnaire is a general client-experience instrument, not a validated explainability scale. It contains no technical/non-technical profile and no Likert items mapped to explainability dimensions. We therefore report its ordinal variables descriptively and

do not use it to answer the profile-comparison component of RQ1. A second construct threat is the explainability coding framework. Labels are multi-valued and the study did not preregister a closed universe of dimensions. We report observed label frequencies but do not calculate a framework-coverage percentage that lacks a defensible denominator.

6.4 Conclusion validity

The strict code-level saturation indicator is 6.306%, above the 5% threshold; declaring saturation would overstate the evidence. Axial stabilization is reported separately as supportive context. For the profile effect size, the corrected independent unit is the session (three per profile), yielding a wide primary 95% interval that includes zero; consequently, the result is explicitly marked non-interpretable for population inference. The design also lacks a preregistered participant-level quantitative endpoint for technical versus non-technical walkthroughs and lacks two equivalent rater rounds for Fleiss' κ . Therefore, Shapiro–Wilk, Levene, Mann–Whitney, and Fleiss tests are not performed. This reduces inferential breadth but avoids p-hacking and HARKing risks from constructing analyses after observing the data Simmons et al. [2011], Kerr [1998].

Table 5: Principal validity threats, mitigations, and residual limitations.

Category	Threat	Mitigation	Residual limitation
Internal	Walkthroughs predate OSF registration.	Explicit preregistration deviation; descriptive framing.	Timing cannot be changed retroactively.
Internal	No member-checking audio/video.	Preserve dated documentary decisions; do not fabricate recording.	Literal recording expectation remains unmet.
External	One local-gym case.	Bound claims to FabroGym and similar contexts.	No statistical generalization.
External	Three sessions per profile.	Effect size uses sessions as independent units and is explicitly descriptive.	Larger replication needed before population interpretation.
Construct	Questionnaire is not an explainability scale.	Keep ordinal items descriptive.	Explainability satisfaction cannot be estimated.
Construct	No closed explainability-dimension denominator.	Report multi-label frequencies only.	Framework-coverage percentage undetermined.
Conclusion	Code saturation indicator is 6.306% (>5%).	Report exact value and axial result separately.	Strict code saturation not established.
Conclusion	Profile effect based on only 3+3 independent sessions; primary CI is wide and crosses zero.	Report $n_{units} = 6$, exact-bootstrap CI and interpretable=NO ; do not use categories as observations.	No inferential/population profile effect.

7 Conclusions and future work

For **RQ1**, the walkthrough evidence shows that explainability needs in the FabroGym case center on four forms of information: the objective and relevant constraints behind a routine, human review and professional provenance, routine history, and the principal criterion behind a future suggestion or automatic assignment. Nine fragments supported this subset, with more explainability-relevant material in technical sessions (7) than non-technical sessions (2). The corrected session-level effect estimate (Cliff's delta=0.556; 95% exact-bootstrap CI [-0.333, 1.000]; 3+3 sessions) remains descriptive and is explicitly non-interpretable as a population effect.

For **RQ2**, those needs were converted into four measurable NFR candidates and

reviewed by MC-P01–MC-P03. Across 12 decisions, four were confirmations and eight were adjustments; no candidate was rejected. The final RNF are more conservative than the candidates because member checking removed unsupported comparisons and overly broad content. All four have a metric, threshold, and verification method, but all remain specified requirements for a proposed recommender rather than implemented AI functionality.

The consolidated contribution is a transparent requirements-engineering pathway from field evidence to operationalized explainability NFRs in a small-organization context. Equally important, the study documents what the evidence does *not* support: no explainability Likert score, no Mann–Whitney or Fleiss result, no claim of strict code saturation, and no claim that a recommendation model has been implemented. These boundaries make the case reproducible and falsifiable.

Future work has three concrete directions. First, additional walkthroughs should test whether the code-level curve falls below the prespecified 5% threshold and whether profile-specific patterns persist. Second, a dedicated explainability instrument should collect dimension-level ratings from technical and non-technical users, enabling preregistered inferential comparisons and confidence intervals. Third, after implementing a routine recommender, the four RNF should be converted into executable acceptance tests and evaluated across alternative explanation interfaces and recommendation architectures, connecting RE specification to system-level explainability evaluation Chazette et al. [2022], Gedikli et al. [2014], Zhang and Chen [2020].

8 Retrospectiva del examen suspenso / Failed-exam correction retrospective

The corrective pass was treated as an auditable closure exercise rather than as an opportunity to rewrite the historical record. Historical authorship remains preserved in Git, the ERS/SRS, and the published research artifacts. The current closure/examination team is Mera Arias Erick Jhair, Mora Duarte Alex José, and Ponce Rivera Mery Helenmey. Alvia Villegas Erick Adalberto and Vaca Romero David Octavio remain credited for their historical contributions where those contributions actually occurred, but they are not presented as pending A2 members of the current closure. The canonical scope statement is `10_Autoria/EQUIPO_EXAMEN_FINAL.md`.

Table 6: Corrections performed after the failed-exam review and documented responsibility.

Item	Review finding	Terminal correction and evidence	Documented responsibility
§15	Closure documentation had mixed historical five-person authorship with the current examination team; the photographic inventory also required normalization.	Historical authorship is preserved, while the current closure/examination team is explicitly Mera, Mora, and Ponce. A2 documents 13 captures for Mera, 10 for Mora, and 19 for Ponce. The EXIF inventory contains 11 real records (2 team, 4 environment, 5 questionnaire).	Repository-side correction completed; no A2 pending item is assigned to historical participants outside the current closure team.
§7	Four public consent PDFs did not themselves evidence a scanned signature because the prior representation was Word-generated; the guide also reports 17 consents while the reproducible 16-session inventory contains 16 individual forms.	ENTR-02, ENTR-03, ENTR-04, and ENTR-06 were replaced by rasterized, redacted scans. The 16-form inventory is documented without fabricating a seventeenth consent, and a byte-identical mirror in 08_Etica/consentimientos / supports the guide’s literal verification command.	Mera Arias / Emeraxs (repository correction record a68594a); count reconciliation documented at closure.
§13	The previous effect-size calculation treated thematic categories as if they were independent observations.	The independent unit is now the WALK session: three technical and three non-technical sessions (n_unidades=6). The primary Cliff’s δ is 0.555556 with exact-bootstrap 95% CI [-0.333333, 1.000000], and interpretable=NO for population inference. The questionnaire $n = 70$ is explicitly excluded from the technical/non-technical explainability contrast because it has neither a profile variable nor explainability items.	Mora Duarte / amorad35 (repository correction record 01c87fb); terminal table/report synchronization reviewed by the closure team.
§16	The report did not consolidate the corrected effect table, survey limitation, consent correction, closure baseline, and a named correction retrospective in one auditable artifact.	This source and its generated PDF now consolidate those elements, including Table 3, the validity limitations, the consent correction above, this retrospective, and the distinction between the historical published baseline and the terminal closure tag declared for this delivery. The report is regenerated from 07_Publicacion/manuscrito_final.tex with 07_Publicacion/compilar_manuscrito.py.	Cross-review of the corrective report; this responsibility statement does not alter historical authorship.

Declared closure baseline. The canonical repository for the final examination delivery is https://github.com/gleiston-guerrero/FabroGym_ISR401. The annotated tag v2.0.2-final is an already-published historical baseline and is not moved or overwritten. The terminal closure identifier declared for this examination delivery is v2.0.3-final. Following the closure order, that annotated tag is created only after the final content commit and after both SHA-256 manifests have been regenerated and verified. Once published on that final commit, v2.0.3-final constitutes the current closure baseline.

Integrity rule for the retrospective. The retrospective records only evidence that exists in the repository or is generated reproducibly from it. It does not create missing screenshots, signatures, participant responses, statistical observations, or historical contributions. Where the review exposed a limitation that cannot be repaired retroactively—for example, only three WALK sessions per profile or the questionnaire’s lack of explainability/profile variables—the limitation is reported explicitly rather than reclassified as satisfied evidence.

Data and materials availability

The canonical examination repository is https://github.com/gleiston-guerrero/FabroGym_ISR401. The tag `v2.0.2-final` is retained as the published historical baseline. The terminal closure identifier declared for this examination delivery is `v2.0.3-final`; it is created only after the final content commit and successful verification of the root and `07_Datos` SHA-256 manifests. The preregistration is publicly identified at OSF (DOI: 10.17605/OSF.IO/62YSC). The replication package was published on Zenodo as version 2.0.0 (DOI: 10.5281/zenodo.22237884). The later identifier normalization in the repository is not claimed to be byte-identical to that published snapshot. Public materials include anonymized transcripts, the anonymized questionnaire, requirement corpora, traceability, processed results, and reproducible scripts. Identifiable audio/video and original consent materials remain restricted and are not part of the open dataset. Data publication follows FAIR and software-citation principles Wilkinson et al. [2016], Smith et al. [2016], Di Cosmo et al. [2020].

Use of AI-assisted technologies

OpenAI ChatGPT (GPT-5.6 Sol) was used under human review for language polishing, consistency checking, LaTeX/document assembly, and cross-checking the manuscript against the versioned project artifacts. The ChatGPT product session did not expose a user-configurable sampling temperature; therefore, no numerical temperature value is reported. The tool was not used as a substitute for primary evidence, to create participants or interviews, to fabricate statistical results, or to invent references. All numerical claims in the Results section are derived from versioned project data and reproducible analysis outputs. The proposed recommendation/AI component of FabroGym is not presented as implemented.

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Author ORCID identifiers

Erick Adalberto Alvia Villegas: 0009-0001-3777-470X; Erick Jhair Mera Arias: 0009-0001-0068-1796; Alex José Mora Duarte: 0009-0000-2494-2842; Mery Helenmey Ponce Rivera: 0009-0006-6041-9198; David Octavio Vaca Romero: 0009-0000-4457-3095.

Declarations

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Ethics and consent. The project maintains public and restricted evidence zones. ENTR-02, ENTR-03, ENTR-04, and ENTR-06 are represented in the public layer by rasterized, redacted scans of their signed forms; their direct identifiers/signatures remain protected according to the public/restricted separation. No editable draft is represented as proof of signature.

Consent for publication. Only anonymized or pseudonymized research outputs are intended for open publication.

A Reproducibility map

Table 7 links the main numerical claims used in the manuscript to the versioned input and the generated CSV that supports the claim. The unique canonical entry point for the current academic delivery is `07_Datos/scripts/run_all.py`; `06_Experimento/` is retained for provenance and historical traceability. Publication-facing tables in `07_Publicacion/` are synchronized copies, not a second analytical chain. This appendix is an audit aid rather than a new analysis.

Table 7: Reproducibility map for numerical claims in the manuscript.

Claim	Versioned input	Reproducible output / stage
16 sessions; 10 ENTR + 6 WALK	<code>sesiones_multimedia_desde_ficha_v3_1.csv</code>	<code>tabla09_cumplimiento_multimedia.csv</code> ; <code>analyze_multimedia</code> .
Video 06:18:08; audio 06:18:14	Same technical-sheet subset	<code>tabla09_cumplimiento_multimedia.csv</code> ; duration conversion and summation.
Questionnaire $n = 70$	<code>encuesta_clientes_anonizados.csv</code>	<code>tabla06_encuesta_frecuencias.csv</code> ; <code>analyze_survey</code> .
76 coded fragments (49 technical, 27 non-technical)	<code>codificacion_walkthroughs.csv</code>	<code>tabla04_comparacion_perfiles.csv</code> ; <code>analyze_walkthroughs</code> .
Session-level profile effect: $n = 6$, $\delta = 0.556$, v 95% CI $[-0.333, 1.000]$, <code>interpretable=NO</code>	<code>codificacion_walkthroughs.csv</code>	<code>07_Datos/resultados/tablas/tabla_efecto_perfiles.csv</code> ; <code>calcular_efecto_perfiles.py</code> .
9 explainability-relevant fragments	Same coding file + applicability flag	<code>tabla10_dimensiones_explicabilidad.csv</code> ; <code>analyze_explainability</code> .
4 terminal RNF with metric, threshold and verification	Candidate RNF + member-checking records	<code>tabla01_rnf_explicabilidad_final.csv</code> .
12 member-checking decisions (4 confirm, 8 ad-just, 0 reject)	<code>member_checking_estructurado.csv</code>	<code>tabla02_member_checking_por_rnf.csv</code> ; member-checking summary.
Strict code indicator = 6.306%	New-code curve across six WALK sessions	<code>tabla03_resumen_saturacion.csv</code> ; final-three-session rule.
Axial stabilization = 1.852%	Axial-category stabilization source	<code>tabla03_resumen_saturacion.csv</code> ; complementary indicator.
Privacy index mean 3.971, 95% bootstrap CI 3.700–4.214	Questionnaire privacy item	<code>tabla05_encuesta_indices_ordinales.csv</code> ; 10,000 bootstrap resamples.
Power references $n \approx 34$ $d = 0.5$, $\alpha = .05$, power = .80 and $n \approx 64$ /group		<code>tabla08_power_calculation.csv</code> ; Statsmodels power functions.

B Analysis applicability decisions

The analysis plan was constrained by the instrument actually administered. Table 8 records the decision rule for statistics explicitly mentioned in the terminal rubric but unsupported by the collected variables. This log prevents later reinterpretation of general questionnaire items as explainability measures.

Table 8: Applicability log for statistical procedures.

Procedure	Status	Evidence-based reason
Explainability Likert mean by dimension + 95% CI	Not applicable	The real questionnaire has no Likert items mapped to explainability dimensions. General ordinal indices are reported only as domain descriptors.
Fleiss' κ	Not calculated	There are not two equivalent rounds of categorical rating of the same units by multiple raters; member checking records confirm/adjust/reject decisions.
Mann–Whitney U , technical vs non-technical	Not applicable	A terminal session-level effect descriptor exists, but there is no preregistered participant-level quantitative endpoint; only three sessions exist per profile.

Table 8: Applicability log for statistical procedures (continued).

Procedure	Status	Evidence-based reason
Cliff’s delta + exact-bootstrap 95% CI by WALK session	Applied descriptively	Six independent sessions (3+3); primary $\delta = 0.556$, CI [-0.333, 1.000]. The output marks <code>interpretable=N0</code> for population inference.
Shapiro–Wilk and Levene tests	Not applicable	No inferential hypothesis on a quantitative outcome is executed for the Enfoque 3 profile comparison.
Bootstrap CI for general ordinal questionnaire dices	Applied descriptively	A fixed-seed 10,000-resample bootstrap summarizes six general ordinal questions; these intervals are not interpreted as explainability effects.

C Candidate-to-final requirement audit

The candidate-to-final mapping in Table 9 makes visible where member checking changed analyst wording. The three reviewers are retained only under the pseudonyms MC-P01, MC-P02 and MC-P03.

Table 9: Member-checking audit by candidate RNF.

Candidate → final	Confirm	Adjust	Reject	Final decision
RNF-EXP-C01 → RNF-16	2	1	0	Retained with narrower wording on objective/need and relevant recorded constraints.
RNF-EXP-C02 → RNF-17	2	1	0	Retained with trainer-review status and external-professional provenance when recorded.
RNF-EXP-C03 → RNF-18	0	3	0	Retained after removing automatic comparison; history, dates and available observations remain.
RNF-EXP-C04 → RNF-19	0	3	0	Retained after removing alternative comparison; only the principal suggestion/assignment criterion remains.

D Manuscript-facing data dictionary

Table 10 documents the variables most directly used to obtain the results in this paper. It is intentionally limited to analysis-facing fields and does not include direct personal identifiers.

Table 10: Data dictionary for the principal manuscript-facing variables.

Variable	Source	Meaning in this study
Codigo_Sesion	Walkthrough coding	Stable identifier: WALK-TEC-01..03 or WALK-NTEC-01..03.
Perfil	Walkthrough coding	Technical or non-technical walkthrough profile; descriptive grouping only.
Categoria	Walkthrough coding	Axial/thematic category assigned to a coded fragment.
Codigo_Normalizado	Walkthrough coding	Normalized open code used for the code-accumulation curve.
Aplicable_Explicabilidad	Walkthrough coding	Flag identifying fragments pertinent to explainability requirement elicitation.
Dimension_Explicabilidad	Walkthrough coding	One or more observed explainability labels; multi-label and not a closed framework denominator.
privacy_importance	Client questionnaire	General ordinal privacy-importance item; not an explainability scale.
Metrica, Umbral	Candidate RNF	Operational measure and acceptance threshold for a candidate/final explainability RNF.
Metodo_Comprobacion Resultado	Candidate RNF Member checking	Verification procedure associated with the RNF. Participant decision: Confirmado, Ajustado, or No confirmado.

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